

# CONCEPT NOTE

*IBM SkillsBuild AI Strategy & Business Intelligence Internship | AICTE*

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**Project Title:** BusSync — AI-Powered Campus Transport Management System

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## 1. Background / Context

College transport systems in India still largely rely on manual communication. The bus route changes due to merges, driver unavailability, or stop changes, but students and staff have no reliable way to get this new information. Drivers or conductors need to be asked directly, confirmation will be needed or buses might be missed. This daily inefficiency messes up academic schedules and causes unnecessary stress for thousands of commuters. The problem is not confined to one AICTE-affiliated college in India but is widespread. The absence of a real-time, digitally managed transport update mechanism is a critical gap in institutional infrastructure — one that has direct implications for SDG 4 (Quality Education), SDG 9 (Industry, Innovation & Infrastructure) and SDG 11 (Sustainable Cities & Communities).

## 2. Objectives

- Build a web/mobile platform that allows the college transport in-charge to update bus numbers, merged routes, and changed stops every morning.
- Deploy an AI agent (powered by IBM Watsonx / OpenAI) that autonomously monitors route data, detects changes, and sends real-time WhatsApp alerts to students and staff.
- Enable a student-facing chatbot that answers route queries instantly using live Google Sheets data.
- Eliminate manual dependency on drivers and conductors.

## 3. Target Audience / Beneficiaries

### Primary Beneficiaries:

- Students (daily commuters who rely on college buses for campus access)
- Faculty and staff using bus facilities

### Secondary Beneficiaries:

- Transport in-charge and college administration (reduced helpdesk load, streamlined operations)
- Bus drivers and conductors (fewer verbal queries from passengers)

## 4. Key Activities

- Design and develop the BusSync admin panel for transport in-charge to input daily route updates via Google Sheets.
- Build Flow 1 in n8n: Scheduled AI agent that checks Google Sheets every morning, detects route changes using an OpenAI model, and sends WhatsApp alerts via Twilio to all registered students.
- Build Flow 2 in n8n: Student query chatbot that receives WhatsApp messages, queries live route data, and replies with accurate bus information using an AI agent with chat memory.

- Deploy QR codes at bus stands linking students to the live schedule web page.
- Test and validate the system with real route data to demonstrate accuracy and response time.

## 5. Expected Outcomes

- 100% of registered students receive WhatsApp alerts within minutes of a route change being entered by the transport in-charge.
- 70% reduction in time spent by the admin on manual communication of route updates.
- Measurable decrease in transport-related complaints and missed buses.
- A working AI agent prototype demonstrating real-world application of IBM Watsonx and n8n automation.
- Scalable model ready to be deployed across other AICTE-affiliated colleges.

## 6. Timeline

- Phase 1 — Research & Planning (Week 1): Problem validation, SDG mapping, BMC and Lean Canvas, system architecture design.
- Phase 2 — Agent Development (Week 2–3): Build Flow 1 (alert agent) and Flow 2 (chatbot) on n8n. Connect Google Sheets, OpenAI, and Twilio WhatsApp.
- Phase 3 — Testing & Refinement (Week 4): Test with sample route data, refine AI prompts, validate alert delivery and chatbot accuracy.
- Phase 4 — Documentation & Submission (Week 5–6): Prepare concept note, presentation, demo video, and final submission package.

## 7. Budget Estimate

- n8n Cloud (free tier): ₹0
- OpenAI API (free credits for demo): ₹0
- Twilio WhatsApp Sandbox (free trial): ₹0
- Google Sheets (free): ₹0
- Production deployment (server/hosting): ₹500–₹1,500/month (estimated)
- Optional SMS gateway (for non-WhatsApp users): ₹2,000/month (estimated)

## 8. Conclusion

BusSync solves a problem that thousands of college students in India face every day. It combines an AI agent with a simple data entry interface to turn a chaotic manual process into a smart, self-updating system that keeps every commuter informed. The project also directly supports SDG 4, SDG 9 and SDG 11 and demonstrates the practical application of IBM SkillsBuild learnings in AI strategy and business intelligence. Built on n8n, OpenAI and Twilio, BusSync is more than just a concept, but a working prototype that can be deployed with a clear path to adoption across multiple institutions. Your support for this project will make a real difference to how colleges run and communicate their transport operations – and it will be a difference that lasts.

## 9. Contact Information

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